

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary -

Rows: 3,900

- Columns: 18

- Key Features:

- Customer demographics (Age, Gender, Location, Subscription Status)

- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)

- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.

- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	3900
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	3900
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	3900
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	3900
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	3900
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	3900
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	3900
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	3900
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	3900
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	3900

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 - Created age_group column by binning customer ages.
 - Created purchase_frequency_days column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (

We performed structured analysis in SQL WorkBench to answer key business questions: 1.

Revenue by Gender – Compared total revenue generated by male vs. female customers.

	gender	revenue
▶	Male	157890
	Female	75191

2.High-Spending Discount Users - identified customers who used discounts but still spend above the average purchase amount.

	customer_id	purchase_amount_(usd)
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79
	33	67
	35	91
	37	69

3.Top 5 Product by Rating - Find products with the highest average review ratings.

	item_purchased	avg_rating
▶	Gloves	3.8614285714285725
	Sandals	3.8443750000000003
	Boots	3.8187500000000005
	Hat	3.8012987012987005
	Skirt	3.784810126582278

4. Average Purchase Amount between Standard and Express Shipping.

	shipping_type	avg_purchase
▶	Express	60.4752
	Free Shipping	60.4104
	Next Day Air	58.6312
	Standard	58.4602
	2-Day Shipping	60.7337
	Store Pickup	59.8938

5.Subscribers vs Non-Subscribers - Compared average spend and total revenue across subscription status.

	subscription_status	avg_spend	total_revenue
►	Yes	59.4919	62645
	No	59.8651	170436

6.Discount -Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
►	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

7.Customer Segmentation – Classified customers into New ,Returning and Loyal segments based on purchase history.

	customer_segment	count
►	Loyal	3116
	Returning	701
	New	83

8.Top 3 Products per Category – Listed the most purchased products with in each category.

	category	item_purchased	purchase_count
►	Accessories	Jewelry	171
	Accessories	Sunglasses	161
	Accessories	Belt	161
	Clothing	Blouse	171
	Clothing	Pants	171
	Clothing	Shirt	169
	Footwear	Sandals	160
	Footwear	Shoes	150
	Footwear	Sneakers	145
	Outerwear	Jacket	163
	Outerwear	Coat	161

9.Repeat Buyers and Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status	repeat_buyers
►	Yes	958
	No	2518

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group .

	age_group	total_revenue
►	Young	62143
	Middle-aged	59197
	Adult	55978
	Senior	55763

5. Power Bi Dashboard



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.